

Problem

Most VQA models optimize **only answer accuracy** and reach the answer as a black box — giving no human-readable evidence for *how* the answer was derived.

- **Type 1** — answer only, no supporting evidence.
- **Type 2** — answer plus a generic caption describing the image.
- **Type 3** — answer plus a self-contained explanation that matches it (rare).

*Almost every state-of-the-art VQA model just answers the question, **without justifying** how it got there.*

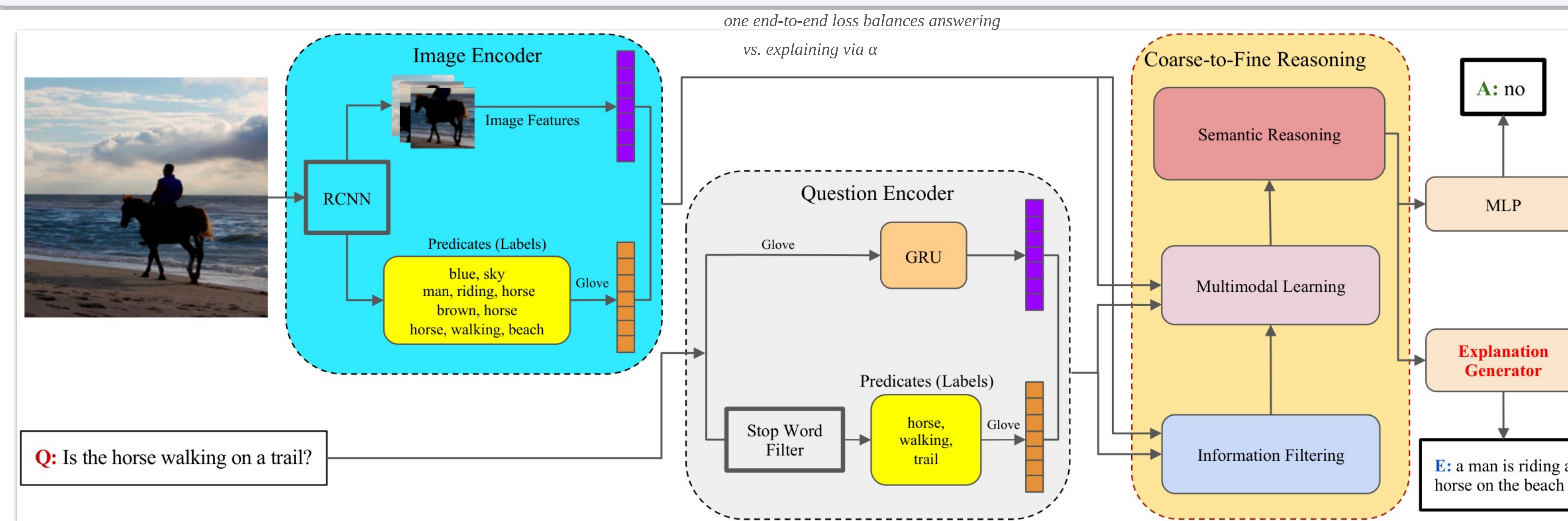
Motivation

- As reasoning demands in VQA grow, **interpretability** becomes urgent for diagnosing and trusting models.
- Explanation quality has **no reliable metric** — BLEU/ROUGE score n-gram overlap, not reasoning.
- Prior explainable-VQA work reuses image captions or external knowledge that gives no direct evidence.
- Two open questions remain: can a model **explain while keeping accuracy**, and how should we reliably **evaluate** those explanations?

Method

- Builds on the **Coarse-to-Fine Reasoning (CFR)** backbone that yields a joint image–question embedding.
- Faster-RCNN region features + a GloVe+GRU question encoder feed the reasoning modules.
 - The joint embedding drives an answer MLP and a new **explanation generator** — an LSTM or Transformer decoder trained end-to-end.

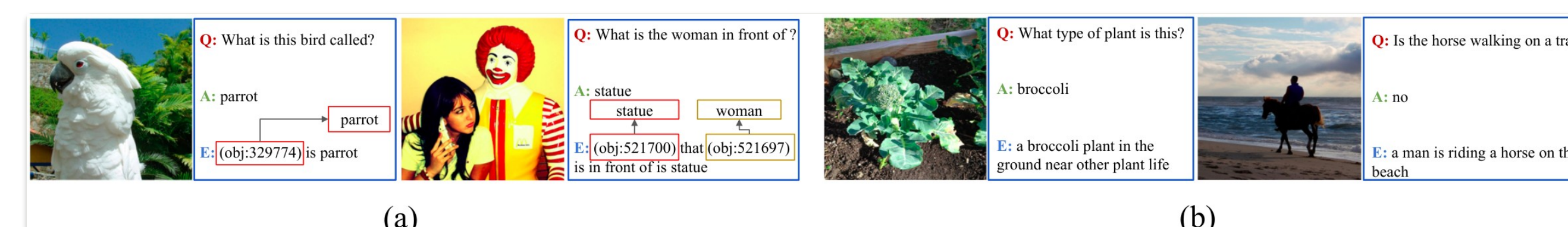
$$L = \alpha L_{ans} + (1 - \alpha) L_{expl}$$



Coarse-to-fine VQA augmented with an end-to-end explanation generator.

Dataset / Benchmark

- **GQA-REX**: ~1.04M QA pairs over 82K images; explanations for ~98% of GQA-balanced samples (scene-graph format).
- **VQA-E**: explanations for ~40% of VQA2.0 QA pairs — caption-like, containing the answer.



Example image–question pairs with annotated explanations from (a) GQA-REX and (b) VQA-E.

Key Results

VQA-E explanation	BLEU-1	ROUGE-L
Prior baseline	0.268	0.249
CFRF + LSTM (ours)	0.33	0.325

Answer accuracy held at 77.49% / 71.48% — explanations added essentially for free.

so WHAT → Explanations improve over prior work while VQA accuracy stays at SOTA — better explanations at **zero** answer-accuracy cost.

Ablation Study

CONFIG	GQA-REX	VQA-E
LSTM (α sweep)	75.1–77.5	71.3–71.6
Transformer ($\alpha=0.5$)	77.06	71.46

Headline Numbers

77.49%
GQA-REX VQA score

71.48% VQA-E VQA score
65.16% explanations reach the answer
~60.5% answer + valid explanation

Takeaway

A simple explanation generator gives a strong VQA model human-readable explanations at **almost no accuracy loss** — but string-matching metrics like BLEU/ROUGE **reward wrong** and **penalize valid** explanations.

The real bottleneck for explainable VQA isn't the generator — it's the evaluation metric.

Questions? Get in touch

rmahesh3@illinois.edu



Paper